

SWB2 Developer Quickstart (pixi)

Last updated: July 2026 **SWB2 version:** v2.4.0-rc0

Overview

SWB2 uses [pixi](#) to manage the build toolchain (gfortran, meson, ninja) and C libraries (netCDF, HDF5, zlib). This eliminates manual dependency installation — everything is downloaded into a local `.pixi/` directory.

From a clean checkout to a working executable:

```
pixi install      # one-time: downloads compiler + libraries (~500 MB)
pixi run setup    # configure the build
pixi run build    # compile
```

Prerequisites

- **Windows 10/11** (x86_64)
- **Git** (for cloning the repository)
- **pixi** (single binary, no admin required)

Install pixi

From PowerShell (no admin):

```
irm get.pixi.sh/install.ps1 | iex
```

Or download manually from <https://github.com/prefix-dev/pixi/releases>

Verify: `pixi --version`

USGS Network Configuration (Required)

USGS uses a TLS-intercepting proxy (MITM) for network monitoring. The pixi binary ships with Mozilla's CA bundle and will **not** trust the USGS proxy's certificate by default. You must create a custom certificate bundle and point pixi at it.

This is a one-time setup per machine. The bundle must contain both the standard Mozilla root CAs (so pixi can reach conda-forge) and the DOI certificate (so pixi trusts the USGS proxy).

Step 1: Create a working directory for certificates

```
mkdir C:\Users\<username>\DOI_Cert
```

Step 2: Export the DOI Root CA from Windows

1. Press `Win+R`, type `certmgr.msc`, press Enter
2. Expand **Trusted Root Certification Authorities** → **Certificates**
3. Find the DOI root certificate (e.g., "DOIRootCA2")
4. Right-click → **All Tasks** → **Export...**
5. Click Next → Select **Base-64 encoded X.509 (.CER)** → Next

- 6. Save to: `C:\Users\<username>\DOI_Cert\DOIRootCA2.cer`
- 7. Click Next → Finish

Step 3: Obtain the Mozilla CA bundle

You need the standard set of public root certificates. Get it from one of these sources:

- **From an existing miniforge/miniconda installation** (if you have one):

```
copy C:\Users\<username>\AppData\Local\miniforge3\Library\ssl\cacert.pem
C:\Users\<username>\DOI_Cert\ca-bundle.pem
```

- **Download from curl.se** (if you have internet access on a non-USGS network): Download `cacert.pem` from <https://curl.se/docs/caextract.html> and save it as `C:\Users\<username>\DOI_Cert\ca-bundle.pem`
- **Bootstrap from pixi itself** (on USGS network, see note below): Temporarily allow pixi to bypass SSL verification to download packages, then grab the bundle from the resulting environment. See Step 3a.

Step 3a: Bootstrap method (if no other source available)

- 1. Create `C:\Users\<username>\.pixi\config.toml` with:

```
tls-no-verify = true
```

- 2. From the swb2 repo directory, run:

```
pixi install
```

- 3. Copy the Mozilla bundle from the pixi environment:

```
copy
E:\projects\swb_development\git\swb2\.pixi\envs\default\Library\ssl\cert.pem
C:\Users\<username>\DOI_Cert\ca-bundle.pem
```

- 4. **Delete** `C:\Users\<username>\.pixi\config.toml` (or remove the `tls-no-verify` line). Do not leave this in place.

Step 4: Append the DOI certificate to the bundle

```
type C:\Users\<username>\DOI_Cert\DOIRootCA2.cer >> C:\Users\<username>\DOI_Cert\ca-bundle.pem
```

This adds the DOI root CA to the end of the Mozilla bundle, creating a single file that trusts both public internet CAs and the USGS proxy.

Step 5: Set the SSL_CERT_FILE environment variable (permanent)

- 1. Press the **Windows key** and type `environment variables`
- 2. Select **"Edit environment variables for your account"** (this does NOT require admin privileges, unlike `sysdm.cpl` which is locked down on managed USGS machines)
- 3. Under **User variables**, click **New**:
 - Variable name: `SSL_CERT_FILE`
 - Variable value: `C:\Users\<username>\DOI_Cert\ca-bundle.pem`

- 4. Click OK
- 5. **Close and reopen** any command prompt windows (the new variable is not picked up by existing shells)

Verify

Open a **new** command prompt and run:

```
echo %SSL_CERT_FILE%
```

It should print `C:\Users\<username>\DOI_Cert\ca-bundle.pem`.

Then test pixi connectivity:

```
cd E:\projects\swb_development\git\swb2
pixi install
```

If this completes without SSL errors, the certificate configuration is correct.

Build Instructions

1. Clone the repository

```
git clone <repository-url> swb2
cd swb2
```

2. Install dependencies

```
pixi install
```

This downloads gfortran, meson, ninja, netCDF, HDF5, and zlib into `.pixi/`. First run takes 5-10 minutes (longer on USGS network due to security scanning of new binaries). Subsequent runs are fast.

3. Configure the build

```
pixi run setup
```

This runs `meson setup` with the correct library paths. On first run this may be slow (~5 minutes) due to endpoint security scanning the new compiler binaries. Second and subsequent runs are much faster.

4. Compile

```
pixi run build
```

Produces three executables in `builddir/src/`:

- `swb2.exe` — main soil-water-balance model
- `swbstats2.exe` — post-processing statistics tool

And in `builddir/test/unit_tests/`:

- `swbtest.exe` — unit test runner

5. Run tests

```
pixi run test
```

Or run directly:

```
builddir\test\unit_tests\swbtest.exe
```

Build Profiles

Command	Profile	Purpose
<code>pixi run setup</code>	release	Optimized, FPE traps, backtrace
<code>pixi run setup-dev</code>	develop	Runtime checks, NaN init, FPE traps
<code>pixi run setup-sa</code>	static_analysis	Maximum warnings, pedantic

After changing profile, re-run `pixi run build`.

How It Works

pixi.toml declares:

- **Compiler:** gfortran 15.x (from conda-forge)
- **Build tools:** meson, ninja
- **Libraries:** libnetcdf, hdf5, zlib

The pixi tasks pass `-Dnetcdf_root=%CONDA_PREFIX%/Library` to meson, which tells it where to find the netCDF/HDF5 libraries. No manual path configuration needed.

Troubleshooting

"SSL certificate problem" or "certificate verify failed" during `pixi install`

You haven't configured the SSL certificate bundle, or it's configured incorrectly. See the USGS Network Configuration section above. Common issues:

- **Forgot to restart your command prompt** after setting `SSL_CERT_FILE` — the env var isn't picked up by existing shells. Open a new one.
- **Bundle doesn't include Mozilla roots** — you appended the DOI cert but didn't start with the Mozilla CA bundle. The bundle needs BOTH.
- **`SSL_CERT_FILE` was set in pixi's config.toml** — this is a Windows environment variable, NOT a pixi config option. Set it via System Properties.

Build is extremely slow (5+ minutes for setup)

This is endpoint security (CrowdStrike/Defender) scanning new unsigned binaries on first encounter. It should be fast on subsequent runs. If it remains slow, ask IT to add exclusions for your project directory and `.pixi/` folders.

Link errors mentioning `HDF5::HDF5` or `CURL::libcurl`

This means the build accidentally used pkg-config's broken netcdf.pc file instead of the `find_library` path. Ensure you're using the current `src/meson.build` which uses `find_library` directly.

"netcdf not found" or "hdf5 not found"

The `CONDA_PREFIX` environment variable isn't being expanded. Make sure you're running via `pixi run setup` (not calling meson directly).

Want to build without pixi?

The build still supports manual configuration:

```
meson setup builddir -Dnetcdf_root="C:/path/to/netcdf"
meson compile -C builddir
```

You'll need meson, ninja, gfortran, and the netCDF/HDF5 libraries installed manually.

Distributing the Executable

The pixi build produces a dynamically-linked executable. To run `swb2.exe` on another machine, copy these DLLs from `.pixi/envs/default/Library/bin/` alongside the `.exe`:

- `netcdf.dll`
- `hdf5.dll`
- `hdf5_hl.dll`
- `zlib.dll` (or `zlib1.dll`)

The Fortran runtime is statically linked (`-static-libgfortran`) so no Fortran DLLs are needed.

Quick Reference

```
pixi install          # install/update dependencies
pixi run setup        # configure (release)
pixi run setup-dev    # configure (develop, with runtime checks)
pixi run build        # compile
pixi run test         # run unit tests
pixi run clean        # remove builddir
```